

Vikash Kumar

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SUMMARY

Computer Engineer and Founder with 1+ year of experience shipping AI-assisted products. Skilled in autonomous systems, machine learning frameworks, and cross-functional coordination. Seeking roles in Product, Operations, or Technical Program Management to leverage fast learning and end-to-end execution capabilities.

SKILLS

- Technical:** Python, React Native, Streamlit, Docker, Git, n8n, Supabase, AI-Assisted Development (Cursor, Claude Code), Web Development, Excel, Data Reporting.
- Product & Strategy:** Product Thinking, Feature Ownership, Roadmapping, SEO Optimization, Execution.
- Management:** Team Leadership, Cross-Functional Coordination, Stakeholder Communication, Decision Making.
- Core:** Problem Solving, Fast Learning, Ownership Mindset, Digital Motion Design.

EXPERIENCE

Founder 2025 – Present
AI Agency

- Direct an AI agency delivering technical solutions and automation pipelines for clients.
- Architect and maintain self-hosted automation workflows utilizing Docker, n8n, Python, and Supabase.
- Explore operational implications and system designs leveraging large language models and machine learning frameworks to drive business value.

AI-Assisted Developer Jan 2024 – Jan 2025
Techype Pvt. Ltd. | Noida, UP, India

- Built an end-to-end talent acquisition platform using AI-assisted tools (Cursor, Claude Code), reducing time-to-hire for early users by streamlining candidate sourcing and screening workflows.
- Improved on-page SEO and content strategy, driving a measurable increase in organic platform visibility and inbound traffic.
- Owned the complete feature lifecycle from ideation and scoping through development, testing, and deployment in a fast-moving startup environment.
- Coordinated a small cross-functional team to ensure timely delivery of product milestones; acted as the bridge between technical and business priorities.

IT Manager Jul 2024 – Jan 2025
Aparc Healthcare | New Delhi, India

- Managed IT infrastructure and internet devices across a multi-department healthcare facility, ensuring 99%+ uptime for critical operations.
- Assisted doctors and staff in bridging technology gaps, improving daily operational efficiency through better tool adoption.
- Supported patient data management, scheduling systems, and reporting workflows using Excel and internal tools.
- Maintained and troubleshooted the organization's website, ensuring accurate information and smooth user experience for patients.

Physics Tutor (Part-Time) 2023 – 2024
Independent

- Taught Physics to high school students, developing strong skills in breaking down complex technical concepts into clear, actionable explanations.

PROJECTS

KV-Cache Compression & Optimization Prototype

- Engineered a mathematical simulation in Python demonstrating token-eviction strategies and PagedAttention logic to reduce memory bottlenecks during long-context LLM generation.
- Solved significant LLM inference constraints by modeling computational complexity and VRAM optimization techniques.

Direct Preference Optimization (DPO) from Scratch

- Implemented the mathematical DPO loss function in PyTorch, fine-tuning a local Llama-3-8B model to align outputs without relying on traditional RLHF frameworks.
- Orchestrated advanced fine-tuning architectures to demonstrate mastery of modern model alignment mathematics.

Custom GPU Kernel via OpenAI Triton (Attention Optimization)

- Authored a custom Sliding Window Attention mechanism using Triton, benchmarking its memory footprint and throughput against standard PyTorch implementations.
- Optimized VRAM usage and memory bandwidth, proving deep understanding of hardware constraints and numerical stability.

EDUCATION

B.Tech in Computer Science and Engineering (CSE) Aug 2016 – Jul 2020
GB Pant Govt. Engineering College (GGSIPU)

CERTIFICATIONS & COURSEWORK

- Relevant Coursework:** Advanced Coursework in AI/ML | MIT OpenCourseWare & IIT (Self-Paced) | 2020 – 2023
- Certifications:** Master of Artificial Intelligence (UPenn) via Coursera, Docker Containerization, AI-Assisted Development, Data Analysis Using PySpark, Google Data Analytics Professional Certificate